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MILITARY SPECIFICATION

CRYSTAL UNITS, QUARTZ, WIRE-MOUNTED, METAL PLATED

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers crystal units consisting of metal-plated quartz oscillators or resonators, wire mounted in sealed holders and suitable for operation over the temperature range of -40°C to +70°C.

1.2 Classification. The crystal units shall be of the following types and frequency ranges:

Crystal unit type	Frequency range
Crystal Unit CR-2D/U	200 kHz
Crystal Unit CR-2D/U	500 kHz
Crystal Unit CR-3C/U	300 to 600 kHz
Crystal Unit CR-4C/U	500 to 1200 kHz
Crystal Unit used in AN/FRR-3( )	462.45 kHz

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder; Soft (Tin, Tin-Lead, and Lead Silver).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-H-10056 - Holders, Crystal.
MIL-H-10056/8 - Holder (Enclosure), Crystal, Type HC-17/U.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ 07703-5601, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electric Component Parts.
MIL-STD-129 - Marking for Shipment.

HANDBOOK

H4/H8 - Commercial and Government Entity (CAGE).

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-512 - Standard Methods for Measurement of the Equivalent Electrical Parameters of Quartz Crystal Units, 1kHz to 1GHz.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20005.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC-444-1 - Measurement of Quartz Crystal Unit Parameters by Zero Phase Technique in a π -Network.

Part 1: Basic Method for the Measurement of Resonance Frequency and Resonance Resistance of Quartz Crystal Units by Zero Phase Technique in a π -Network.

(Application for copies should be addressed to the American National Industries, 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Parts, materials, and processes; general.

3.1.1 Selection. Items (parts, materials and processes) shall be such as to enable the equipment to meet its performance and other requirements as described herein, and shall be selected in accordance with subparagraphs as applicable.

3.1.1.1 Selection criteria. The item shall conform to a detailed document referenced herein for the particular purpose of specifying the item; to any changes in, or additional requirements prescribed by, this specification in connection with the detailed document, and to any limitations prescribed herein as to application of types, classes, grades, characteristics, composition, etc., covered by the detailed document. Optional deviations from the detailed document, if described herein, may be exercised by the contractor.

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3.2 First article. Crystal units furnished under this specification shall be products which have been tested and passed the first article inspection specified in 4.3.

3.3 Soldering.

3.3.1 Solder. Solder used for electrical connections shall be composition SN60 conforming to QQ-S-571. When the design of the crystal unit involves the use of a nonferrous metal in direct contact with the plating, a tin-lead eutectic solder saturated with the metal used for plating the quartz surfaces shall be employed. Saturation of the solder with the plating metal shall be such that during soldering, or thereafter, migration of metal from the plated surface of the quartz plate to the solder will be effectively checked. Springs shall be attached to the pins by using a high temperature solder or by welding.

3.3.2 Flux and cleaning agents. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin or rosin and alcohol.

3.3.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. The solder shall be so accomplished that no rosin remains on the joint or in the proximity of the joint. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4 Insulating, impregnating, and sealing compounds. No insulating, impregnating, or sealing compounds shall be used which, either in the state of their original application or as a result of aging, may cause any injurious effect upon any part of this unit. No material whatever, unless it has been specifically approved for the purpose, shall be placed around the base of the pins after they have been assembled in the holders.

3.5 Design and construction.

3.5.1 Quartz crystal element. The quartz crystal element shall be wire-mounted and shall have its major faces plated with a metallic film.

3.5.2 Crystal holder. Crystal holder HC-17/U shall be used for each crystal unit. The physical dimensions and materials shall conform to all requirements and tests of MIL-H-10056 and MIL-H-10056/8. The length and width of the crystal unit after solder sealing shall not exceed the maximum length and width specified for the unassembled crystal holder by more than .007 inch (0.18 mm) in each case.

3.5.3 Reference standard test sets. The reference standard test set used for calibration of all the crystal units shall be the Crystal Test Set AN/TSM-2( ) or its equivalent. If the contractor uses an equivalent reference standard test set it shall conform to IEC-444 or EIA-512. The contractor shall be responsible for certifying that the equivalent reference standard test set conforms to the appropriate standard (see 3.5.3.1). The reference standard test set shall be used to check the characteristics and accuracy of the contractor's equivalent test set.

3.5.3.1 Certification of reference standard test sets. Certification of compliance with IEC-444 or EIA-512 shall consist of the following:

a. Functional diagram of the contractor's system showing interconnections, equipment manufacturer, and model designations.

b. Flow charts and descriptions of all software modules used in the control of equipment and estimation of parameter values.

c. Flow charts and descriptions of instrument calibration and error correction routines.

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d. Traceability documentation for test fixtures and reference impedance devices required for the calibration, verification, and use of the standard test set.

3.6 Temperature. Crystal units covered by this specification shall operate satisfactorily over the temperature range of -40°C to +70°C ±2°C.

3.7 Frequency. At no temperature within the above temperature range shall the frequency of the unit deviate from the nominal more than the amount permitted for each individual unit, when tested in accordance with 4.7.1 and 4.7.2.

3.7.1 Frequency drift. During the test of 4.7.4.1, the frequency drift of the unit during the first 30 seconds of operation shall not exceed the amount permitted in the individual requirements. If the frequency drift during the first 30 seconds of operation exceeds the permitted value, the drift during the next 60 seconds shall not exceed the tolerance permitted in the individual requirements.

3.8 Activity. At no temperature within the temperature range shall the activity of the unit fall below the minimum in the applicable requirement for the individual unit, when measured in accordance with 4.8.1 and 4.8.2.

3.9 Vibration. As a result of the test of 4.9, there shall be no resultant mechanical damage to the unit, and its activity shall not have changed more than the respective amounts permitted in the applicable individual requirements paragraphs.

3.10 Seal. During the test specified in 4.10, there shall be no evidence of leakage of gas or air from the inside of the crystal unit.

3.11 Immersion (see 4.11). After the test of 4.10, there shall be no resultant mechanical damage to the unit, and the marking shall be legible. In addition, the dc resistance between the pins of the unit shall be not less than 500 megohms. The activity and the frequency shall be within the applicable tolerances.

3.12 Resistance to corrosion. As a result of the test of 4.14, there shall be no excessive corrosion.

3.13 Pull. For the test specified in 4.13, the minimum value of breaking strength shall be that specified in the individual requirements for that type crystal.

3.14 Humidity resistance. The humidity test, as described in 4.15, shall neither render the unit inoperative, cause permanent degradation in performance, nor cause excessive corrosion or other damage.

3.15 Plating adherence. A test for plating adherence shall be conducted under the direction of the Government in accordance with 4.16. In case of failure, the plating process shall be corrected before any further quartz plates are processed.

3.16 Crystal unit CR-2D/U (200 kHz). The crystal unit required shall be a metal-plated, DT-cut crystal, wire-mounted, for use as a calibrator, designed to operate at anti-resonance into effective load capacitance of 125 pF (see table I).

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TABLE I. Crystal unit CR-2D/U (200 kHz).

Requirement	Specified value or condition
Holder	HC-17/U
Nominal Frequency	200 kHz
Marking	In accordance with 3.21
Frequency tolerance	±18 Hz
Vibration (frequency change permitted)	±0.002%
Pull	Tensile strength of 475 grams minimum
Frequency drift	Not applicable
Activity	.50 mA (-40°C to +55°C) min. .45 mA (+55°C to +70°C) min.

3.17 Crystal unit CR-2D/U (500 kHz). The crystal unit required shall be a metal-plated, DT-cut crystal, wire-mounted, for use as a calibrator, designed to operate at anti-resonance into an effective load capacitance of 64 pF (see table II).

TABLE II. Crystal unit CR-2D/U (500 kHz).

Requirement	Specified value or condition
Holder	HC-17/U
Nominal frequency	500 kHz
Marking	In accordance with 3.21
Frequency tolerance	-65 Hz and +25 Hz (from nominal)
Vibration (frequency change permitted)	±0.002%
Pull	Tensile strength of 150 grams minimum
Frequency drift	Not applicable
Activity	.50 mA (-40°C to +70°C) min.

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3.18 Crystal unit CR-3C/U. The crystal unit required shall be a metal-plated, CT-cut crystal, wire-mounted, designed to operate into a using circuit similar to the test oscillator (see table III).

TABLE III. Crystal unit CR-3C/U.

Requirement	Specified value or condition	
Holder	HC-17/U	
Nominal frequency	300 to 600 kHz	
Marking	In accordance with 3.21	
Frequency tolerance	±0.02%	
Vibration (frequency change permitted)	±0.002%	
Pull	Tensile strength of 350 grams minimum	
Frequency drift	±10 Hz in 30 seconds	±3 Hz in 60 seconds
Activity	0.20 mA min.	

3.19 Crystal unit CR-4C/U. The crystal unit required shall be a metal-plated, CT-cut crystal, wire-mounted, designed to operate into a using circuit similar to the test oscillator (see table IV).

TABLE IV. Crystal unit CR-4C/U.

Requirement	Specified value or condition	
Holder	HC-17/U	
Nominal frequency	500 to 1200 kHz	
Marking	In accordance with 3.21	
load capacitance	125 pF	
Frequency tolerance	±0.02%	
Vibration (frequency change permitted)	±0.002%	
Pull	Tensile load 500 to 700 Hz. - 250 grams 701 to 1200 Hz. - 150 grams	
Frequency drift	In 30 seconds 500 to 600 kHz - 10 Hz 600.001 to 1200 kHz - 20 Hz	In 60 seconds 500 to 600 kHz - 5 Hz 600.001 to 1200 kHz - 10 Hz
Activity	0.10 mA min.	

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3.20 Crystal unit in diversity receiving equipment AN/FRR-3( ). The crystal unit required shall be a metal-plated, CT-cut crystal, wire-mounted. The crystal unit shall be tested on position number 3 in standard oscillator TS-221/TSM (see table V).

TABLE V. Crystal unit in diversity receiving equipment AN/FRR-3( ).

Requirement	Specified value or condition	
Holder	HC-17/U	
Nominal frequency	462.54 kHz	
Temperature range	+25°C to +45°C	
Marking	(On side of crystal) for use in AN/FRR-EB (On top of crystal) frequency 462.54 kHz	
Load capacitance	125 pF	
Frequency tolerance	+20 Hz	
Vibration (frequency change permitted)	±0.002%	
Pull	Tensile load: 350 grams	
Frequency drift	In 30 seconds - 5 Hz.	In 60 seconds - 5 Hz.
Activity	0.20 mA min.	

3.21 Marking. The specified frequency shall be marked on the crystal unit in accordance with MIL-STD-1285, at the top of the unit. The type designation and the manufacturer's CAGE code shall be marked on the largest side of the crystal unit. The manufacturer's designated CAGE code shall be as listed in H4/H8. The type designation, frequency, and manufacturer's code designation characters shall be not less than .062 inch (1.57 mm) in height and located symmetrically with respect to the center axis of the crystal holder in accordance with MIL-STD-1285.

3.22 Workmanship. Crystal units shall be manufactured, processed, and assembled in a thoroughly workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Quantities. After award of contract, 30 samples or 10 percent, whichever is less, of each type of crystal unit called for in contract or order shall be furnished for performance testing. When several nominal frequencies are specified for a particular type crystal unit, this quantity shall consist of an equal number of units of lowest frequency, highest frequency, and the frequency nearest mid-frequency.

4.3.2. Compliance with specified requirements. First article samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect.

4.3.2.1 Approval of first article samples. Approval of first article samples shall not be construed as a waiver of any specified requirement. When deviation from first article requirements is unavoidable, the first article samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefore, and manner in which it will be corrected in production of the units on contract.

4.3.3 Test data. Each submission of samples shall be accompanied by a certification of the materials used in their fabrication (see 3.1), together with a statement of the manufacturer's test data, indicating the frequency and activity measurements made by the contractor.

4.3.4 First article inspection. First article samples will be subjected to the tests shown in table VI, in the order shown. Unlisted nondestructive inspection on first article samples of equipment, and unlisted destructive or nondestructive inspection on separate first article samples of parts or materials, may be performed to determine compliance with specified requirements.

TABLE VI. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical		4.5
Frequency (over temperature range)	3.7	4.7.2
Activity (over temperature range)	3.8	4.8.2
Frequency drift	3.7.1	4.7.4
Vibration	3.9	4.9
Seal	3.10	4.10
Immersion	3.11	4.11
Internal inspection	3.3 and 3.5	4.12
Corrosion	3.12	4.13
Pull	3.13	4.14
Humidity	3.14	4.15
Plating adherence	3.15	4.16

4.3.5 Failures. If two or more units fail in any one of the first article tests, a new lot of samples, 30 units or 10 percent, whichever is less, may be requested. This should be accompanied by a statement of the changes made to overcome the defects that caused the failure of the preceding samples.

4.3.6 Reference standards. After first article samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used when necessary, to check design features and material requirements, e.g., material of holder, kind of plating, and design of wire mount. They shall also be used as reference standards to resolve any difference of opinion regarding interpretation of requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection, general. Government inspection shall consist of group A and group B inspections as specified in 4.4.1 through 4.4.2.2. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group A and group B inspections. Unless otherwise specified herein, all measurements tests shall be made at 25°C +10°C and -5°C, ambient atmospheric pressure, and ambient humidity. (See also 4.7.3 and 4.9.3.)

4.4.1.1 Inspection lot. A lot shall consist of crystal units of only one type, but may consist of assorted frequencies in a particular type. The size of the lot and the manner in which it is formed (output of one assembly line operator, etc. for a period of time such as shift, day, week) shall be agreed upon by the Government inspector and the manufacturer.

4.4.1.1.1 Group A inspection. Group A inspection shall consist of the inspections specified in table VII, in the order shown. When a crystal unit fails the vibration tests, it shall not be replaced by another unit until completion of the seal test.

4.4.1.1.2 Sampling plan. A sample of parts shall be randomly selected in accordance with table VIII, if one or more defects are found, the lot shall be rescreened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table VIII, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE VII. Group A inspections.

Inspection	Requirements paragraph	Test method paragraph
Subgroup I tests (external)		
Visual and mechanical		4.5
Vibration	3.9	4.9
Seal	3.10	4.10
Frequency drift	3.7.1	4.7.4
Subgroup II tests		
Frequency (over temperature range)	3.7	4.7.2
Activity (over temperature range)	3.8	4.8.2

TABLE VIII. Group A, zero defect sampling plan.

Lot size	Sample size 1/
2 to 50	5
51 to 90	7
91 to 150	11
151 to 280	13
281 to 500	16
501 to 1,200	19
1,201 to 3,200	23
3,201 to 10,000	29
10,001 to 35,000	35
35,001 to 150,000	40

1/ If sample size equals or exceeds lot size, 100 percent inspection shall be performed.

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4.4.1.2 Group B inspection. Group B inspection shall consist of the tests specified in table IX and shall be made on sample units that have been subjected to, and passed group A inspection.

4.4.1.2.1 Sampling plan. A sample of parts shall be randomly selected in accordance with table X, if one or more defects are found, the lot shall be rescreened and defects removed. After screening and removal of defects, a new sample of parts shall be randomly selected in accordance with table X, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

4.4.1.2.2 Disposition of sample units. Sample units that have passed the group B inspection may be delivered on the contract or purchase order if the lot is accepted and the sample units are still within specified electrical tolerances.

TABLE IX. Group B inspections.

Inspection	Requirements paragraph	Test method paragraph
Immersion	3.11	4.11
Internal inspection	3.3 thru 3.5	4.12
Pull	3.13	4.13
Plating adherence	3.15	4.16

TABLE X. Group B, zero defect sampling plan.

Lot size	Sample size 1/
2 to 25	2
26 to 50	3
51 to 90	4
91 to 150	6
151 to 280	7
281 to 500	8
501 to 150,000	9

1/ If sample size equals or exceeds lot size, 100 percent inspection shall be performed.

4.4.1.2.3 Noncompliance. If a sample unit fails group B inspection, the contractor shall immediately investigate the cause of failure, and shall report to the Government inspector the results thereof and proposed corrective action to be taken on the process and affected portion of product. If the Government inspector does not consider the corrective action will enable the equipment to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.5 Visual and mechanical inspection. Parts and components shall be inspected for workmanship, mechanical fit, application of specified treatments, and miscellaneous defects. Wiring, soldered connections, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required.

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4.6 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.7 Frequency.

4.7.1 Method measurement. The frequency of the unit shall be measured in an operating crystal test oscillator which has been correlated with the reference standard crystal test set.

4.7.2 Measurement over the temperature range. Frequency measurements shall be made either continuously or at intervals no greater than 3°C over the temperature range while the temperature of the unit is varied at a rate not exceeding 20°C per minute.

4.7.3 Determination of change. In any test requiring that two measurements of the frequency of a unit be made to determine a change induced by some test condition, care shall be taken that the unit has reached thermal equilibrium, and that the ambient temperature in which the second measurement is made is within +2°C of the temperature in which the first measurement was made.

4.7.4 Frequency drift.

4.7.4.1 Amount of frequency drift (see 3.7.1). The amount of frequency drift of the unit during the first 30 seconds of operation shall be measured while the unit is oscillating at room temperature.

4.7.4.2 Frequency drift tolerance (see 3.7.1). If the frequency drift exceeds the allowable tolerance for the first 30 seconds of operation, the oscillation of the crystal shall be allowed to continue without interruption for an additional 60 seconds of operation and the amount of drift shall again be measured.

4.8 Activity.

4.8.1 Method of measurement (see 3.8). Activity of the unit shall be measured in an operating crystal test oscillator which has been correlated with the referenced standard crystal test set.

4.8.2 Measurements over the temperature range (see 3.8). Activity measurements shall be made concurrently with the frequency measurements specified in 4.7.2, either continuously or at intervals no greater than 3°C over the temperature range while the temperature of the unit is varied at a rate not exceeding 20°C per minute.

4.8.3 Determination of change. In any test which requires that two measurements of activity of the unit be made in order to determine a change induced by some test condition, care shall be taken that the unit has reached thermal equilibrium, and that the ambient temperature in which the second measurement is made is within +2°C of the temperature in which the first measurement was made.

4.9 Vibration (see 3.9). The crystal units shall be rigidly mounted on the horizontal platform of the vibration machine. A simple harmonic motion having an amplitude of .015 inch (0.38 mm) (maximum excursion of .030 inch (0.76 mm)) shall be applied continuously to the platform for 2 hours. The frequency of the applied vibration shall be varied at a uniform rate between approximately 10 and 55 cycles per second, the cycle of a frequency excursion being between 1 and 2 minutes. The units shall be mounted as follows:

a. One-third, with a pin length parallel to the direction of motion.

b. One-third, with the motion perpendicular to the largest face.

c. One-third, with the motion perpendicular to the pin length and parallel to the larger face. The frequency and activity shall be measured before and after the vibration test, using the methods specified in 4.7.1, 4.7.3, 4.8.1, and 4.8.3.

4.10 Seal (see 3.10). Prior to the seal test, the crystal unit shall be held at room ambient temperature for a period of at least 24 hours after final sealing. The unit shall then be immersed in water at a temperature between 90°C and 95°C for 2 minutes. No gas or air leakage shall occur, as indicated by bubbles escaping from within the holder.

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4.11 Immersion (see 3.11). The unit shall be immersed in water heated to a temperature above 90°C and held at this temperature for approximately 1 hour. The unit shall then be removed from the water before the water has been permitted to cool. All surface water shall be wiped off or force air dried, and the unit shall be set aside at room temperature for 1/2 hour. At the end of this test, the dc resistance shall be measured between the pins of the unit and also between each pin and any other external metal part of the unit, using a dc test voltage of 50 to 100 volts, and the frequency and activity shall be measured using the methods specified in 4.3.1 and 4.4.1.

4.12 Internal inspection. The unit shall be disassembled and the interior of the holder shall be examined in detail for compliance with the requirements of 3.3 through 3.5.

4.13 Pull. The unit shall be disassembled and shall be tested for breaking strength by applying a steadily increasing tensile load to the lead wires until breakage occurs.

4.14 Corrosion test. The crystal unit shall be tested in accordance with method 101, test condition B, of MIL-STD-202. After this test, the unit shall be examined for compliance with 3.12.

4.15 Humidity (see 3.14). The crystal unit shall be placed in a humidity chamber that contains water maintained at a temperature of 65°C +5°C by means of heat applied to the water. The crystal unit shall not be immersed in the water. After the 30-day test period, the crystal unit shall be removed from the chamber, wiped off, and set aside for a period of 1 hour at room temperature. The frequency, effective-resistance, and vibration measurements specified in 3.7, 3.8, and 3.9 respectively, shall be made.

4.16 Plating adherence (see 3.15). The plating adherence test shall be performed as follows:

a. Firmly apply a piece of transparent plastic pressure-sensitive tape to the plating of the quartz plate.

b. Remove the tape immediately by lifting one corner and pulling at a right angle to the surface of the quartz.

c. Any evidence of removal of plating, as indicated by bare quartz areas or by flakes of plating appearing on the tape shall be an indication of poor adherence. Visual examination is performed without the aids of magnification.

5. PACKAGING

5.1. Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Crystal units shall clean or be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Crystal units shall dry or be dried in accordance with MIL-P-116.

5.1.1.3 Preservatives. Contact preservatives shall not be used.

5.1.1.4 Unit packs. Crystal units shall be unit packed one each and protected in accordance with the requirements contained in MIL-P-116, method III.

5.1.1.5 Intermediate containers. The use of intermediate container shall be in accordance with MIL-STD-2073-1.

5.1.2 Level C. The level C preservation of microphones shall conform to the MIL-STD-2073-1 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Packing and unitization or palletization. Packing and unitization or palletization shall conform to the requirements of MIL-STD-2073-1 for the level cited. When the quantity of the stock numbered item being shipped level A to a single destination comprises a shipping container whose volume is less than 1 cubic foot (.02832 cubic meter) and weight is less than 25 pounds (11.3398), packing shall be as specified for level B.

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5.3 Marking. In addition to any special marking required by the contract or order, marking for levels A, B, and C shall be in accordance with the requirements of MIL-STD-129.

5.3.1 Civil agencies. When specified in the contract or order (see 6.2), the marking of domestic shipments for civil agencies shall be in accordance with FED-STD-123.

5.4 First article and quality conformance inspections. First article, when required (see 6.2), inspections and test shall be as specified in MIL-P-116. The performance of the visual and dimensional inspections, rough handling tests and leakage tests shall conform to the inspections and tests outlined in MIL-P-116.

5.4.1 Functional requirements.

5.4.1.1 Rough handling test (when specified, see 6.2). When packs have been tested in accordance with MIL-P-116, all material and components comprising each pack shall be free from damage or evidence of displacement which might affect the use of the preservation method or pack. The microphones and associated accessories within the tested packs shall show no visible signs of damage. When specified in the contract (see 6.2), functional tests in accordance with the group A inspection requirements of the commodity specification shall be conducted on those devices subjected to the rough handling test to determine freedom from operational malfunction. The examination of the devices tested under this group A inspection shall be in accordance with the visual and mechanical inspection requirements specified in the commodity specification.

5.4.1.2 Leakage test (when applicable). When a barrier enclosed unit pack has been tested in accordance with MIL-P-116, there shall be no evidence of moisture within the unit pack.

5.4.2 Workmanship. The quality of workmanship shall assure acceptance of the completed preservation, packing and marking in accordance with the inspection specified in MIL-P-116.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These crystal units are intended for use in controlling the oscillator frequency of radio transmitters.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Type required.

d. Whether domestic or overseas shipment is required.

e. First article samples required (see 3.2).

f. Marking and shipping of samples.

g. Place of final inspection.

h. Levels of preservation and packing required (see 5.1 and 5.2).

6.3 Subject term (key word) listing.

Crystal
Frequency
Holder

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-A205)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-12813B(ER)	2. DOCUMENT DATE (YYMMDD) 24 August 1993

3. DOCUMENT TITLE
Crystal Units, Quartz, Wire-Mounted, Metal Plated

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Research Laboratory	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 544-3441 (2) AUTOVON: 995-3441
c. ADDRESS (Include Zip Code) ATTN: AMSRL-EP-RD Fort Monmouth, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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